SUPPORTING ADULT GUIDANCE

# The alternate segment theorem

Circle Theorems · Lesson 7. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

## What this lesson does

This lesson establishes the alternate segment theorem: the angle between a tangent and a chord equals the inscribed angle subtended by that chord in the alternate segment. Learners meet it first as an invariant in a draggable lab (the tangent–chord angle and the far-segment angle stay equal as the chord and the viewing point move), then prove it by constructing the diameter from the point of contact and chaining tangent-perpendicularity, the semicircle right angle, and the same-segment theorem. It is the hardest theorem in the unit to recognise, so the emphasis is on spotting the tangent–chord/inscribed-angle pair and always giving the reason.

Driving Question: *Why does the tangent angle match the angle in the far segment?*

## The mathematics, in plain terms

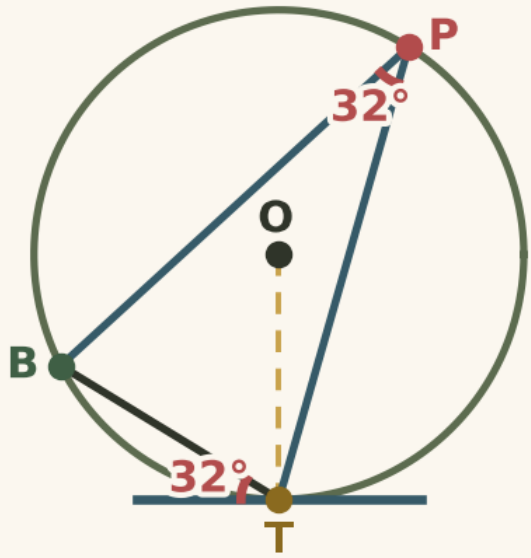
For a tangent touching at T and a chord TB, the tangent–chord angle equals the inscribed angle ∠TPB for any P on the arc on the opposite side of TB. Proof: draw the diameter TT′. Since the tangent is perpendicular to the radius, the tangent–chord angle $x$ gives ∠BTT′ = 90 − $x$. As TT′ is a diameter, ∠TBT′ = 90°, so in triangle TBT′, ∠TT′B = $x$. Finally ∠TPB = ∠TT′B (same segment), so ∠TPB = $x$. The two tangent–chord angles on either side of the chord are supplementary and correspond to the two alternate segments.

## The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

**Example 1 — Copy the angle across the chord**

A tangent touches a circle at T, and chord TB is drawn. The tangent–chord angle is marked. P is a point on the arc on the other side of TB. State the angle in the alternate segment, ∠TPB, and the reason.

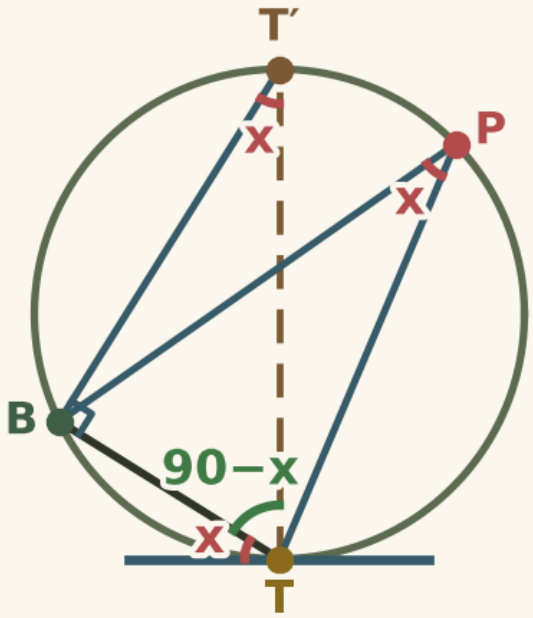



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|---|---|
| **1.** Identify the tangent–chord angle at T and the chord TB. | tangent and chord meet at T |
| **2.** The angle in the alternate segment stands on the same chord from the far arc. | that is ∠TPB at P |
| **3.** The two are equal by the alternate segment theorem. | ∠TPB = tangent–chord angle |

**∠TPB equals the tangent–chord angle — the alternate segment theorem.**

**Example 2 — Why they are equal (the proof)**

Let the tangent–chord angle be $x$. Prove that the angle in the alternate segment, ∠TPB, is also $x$.




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| **1.** Draw the diameter TT′. The tangent ■ the radius, so the tangent–chord angle is $x$ and ∠(tangent, TT′) = 90°. | tangent ■ radius |
| **2.** So the angle between the chord and the diameter is ∠BTT′ = 90 – $x$. | angles at T |
| **3.** TT′ is a diameter, so ∠TBT′ = 90° (angle in a semicircle). In triangle TBT′, ∠TT′B = 180 – 90 – (90 – $x$) = $x$. | semicircle right angle |
| **4.** ∠TT′B and ∠TPB stand on chord TB from the same segment, so ∠TPB = $x$. | same segment |

**∠TPB = $x$ — the angle in the alternate segment equals the tangent–chord angle. ■**

## An invariant first, then a chained proof

Let the learner drag the chord and the viewing point and see the two angles stay equal before naming the rule. Then build the proof one line at a time: the single construction — the diameter from T — is the whole trick, turning the tangent–chord angle into an angle of a right-angled triangle. Reason each step aloud (tangent ■ radius, semicircle right angle, same segment) rather than asserting the result. Insist on naming the theorem when using it. This is the capstone angle theorem; Lesson 8 turns to

the perpendicular from the centre to a chord, and Lessons 9–10 combine all the results in multi-step problems and proofs.

## The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

| Cornerstone | Its role here | Invitational timing |
|---|---|---|
| Connection | See the tangent–chord angle and its twin in the alternate segment. | ~9 min |
| Movement | Copy the equal angle across the chord, with reasons. | ~11 min |
| Reflection | Alternate-segment lab: drag P and the chord; the two angles stay equal. | ~12 min |
| Creativity | Prove it via the diameter, the semicircle right angle, and same segment. | ~10 min |
| Rest | A short pause: a hard result built from facts you already own. | ~2 min |
| Nutrition | Intellectual: perseverance and how proofs are assembled. | ~3 min |

## Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

**Matching the tangent–chord angle to the near segment instead of the alternate one.**
'The equal angle is in the segment on the OTHER side of the chord — the alternate segment.'

**Thinking the angle in the segment changes as P moves.**
'P is in the same segment wherever it sits on that arc, so ∠TPB does not change.'

**Giving a value with no reason.**
'Correct — and the reason is "the tangent–chord angle equals the angle in the alternate segment".'

**Confusing the two tangent–chord angles at T.**
'The two angles either side of the chord add to 180° and match the two different segments.'

## Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

**Please hold this line**
Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

## Reconnection Routes

If an earlier idea is blocking progress, the lesson's Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: what a tangent is and that it meets the radius at 90° (Lesson 6), the angle in a semicircle (Lesson 3), and angles in the same segment (Lesson 4). A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

## Supporting through questions: an example

Suppose a learner matches the tangent–chord angle to the wrong segment, or gives a value with no reason. The aim is a question that surfaces the chord, the far arc, and the reason.

- "Which chord is the tangent angle sitting on? Now look at that chord from the far arc."
- "Is your point P on the same side of the chord as the tangent angle, or the other side?"
- "What is your reason? Name the theorem."
- "If P moves along its arc, does ∠TPB change? Why not?"

If they stall, that is the moment for a Reconnection Route or the lesson's prompt and hint — not for the answer.

## Questions you might have

**What does 'alternate segment' mean?**

A chord splits a circle into two segments. The alternate segment is the one on the opposite side of the chord from the tangent–chord angle you are looking at.

**Why is the tangent–chord angle equal to the angle in the alternate segment?**

Draw the diameter from the point of contact. Tangent-perpendicularity and the semicircle right angle turn the tangent–chord angle $x$ into the angle $\angle TT'B = x$ in a triangle, and the same-segment theorem carries it to $\angle TPB = x$.

**Does it matter where P is on the arc?**

No. Every point on that arc is in the same segment, so $\angle TPB$ is the same for all of them — and all equal the tangent–chord angle.

**How do I spot when to use it?**

Look for a tangent and a chord meeting at the point of contact. The angle between them equals the inscribed angle on that chord seen from the other arc.

## Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

## Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

| Where they are | What it can look like |
|---|---|
| Approaching Proficiency | Beginning to engage; names some features and makes a start with support. |
| Developing Proficiency | Carries out the method with prompts; can explain part of the reasoning. |
| Proficiency | Works through independently and explains why each step is true. |

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|---|---|
| Refining Proficiency | Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else. |

## A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

## If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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